Dynamic Fixed-Batch Retrieval: Structured Codes and Random-Family Locality Barriers

Stamatis-Christos Saridakis

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Abstract

We study a dynamic exact-retrieval problem in which a Boolean vector is maintained under point updates while each query asks for the contents of one set from a fixed family of sparse batches. We call the model Fixed-Batch Dynamic Gather (FBDG). The central question is whether a representation can simultaneously have very low write locality and very low recovery locality when every requested batch has size $w = \Theta(\log n)$ and one dynamic word contains w bits. We prove a structure-versus-randomness separation at write locality two. On the positive side, an explicit Fano-based construction yields infinite families with $M = n$, $w = \Theta(\log n)$, two fixed update locations per source bit, and two adaptive-or-nonadaptive word probes per query. On the negative side, if $M = \Theta(n)$ batches are sampled independently and uniformly from the w-subsets of $[n]$, then with high probability every exact canonical representation in which each source coordinate has at most two fixed update locations requires $q = \Omega(w)$ probes, even when cell contents are arbitrary nonlinear functions and the decoder is fully adaptive. The proof uses a diffuse-versus-hub dichotomy for the fixed support system. Diffuse supports force a linear matching in some random batch uniformly over all support systems. Hub-concentrated supports reduce to a short-sketch problem over bounded residual source groups, where a self-contained adaptive decision-tree counting lemma forces linearly many residual probes. We additionally prove a state-local congestion inequality that applies even when write locations move with the state or history; it shows that any hypothetical $q = o(w)$ moving-support scheme for a random family must maintain high-influence write hubs at every reachable state. Thus the remaining frontier is not generic nonlinearity or adaptivity, but the power of moving support geometry.

1. Introduction

Dynamic data structures are normally specified by an update operation and a query operation, with complexity measured as a trade-off between the two. In many algorithmic and storage settings, however, the query workload is not arbitrary. A fixed collection of small coordinate groups is known in advance, and every query asks for the exact contents of one of those groups. Such workloads arise naturally in batched retrieval, coded storage, materialized views, packed microstructures, and reductions that decompose a larger computation into many fixed local retrieval tasks.

This paper isolates that regime in a minimal exact-recovery problem. The logical state is a Boolean vector

$$x \in \{0,1\}^n,$$

maintained under single-coordinate updates. A fixed family

$$\mathcal{B} = (B_1, \dots, B_M), \qquad B_i \in \binom{[n]}{w},$$

is known at preprocessing time. Query i must return the complete vector $x|_{B_i}$. Dynamic memory is divided into w-bit cells. We measure **write locality** u, the number of dynamic cells that may change after one source-coordinate update, and **query locality** q, the number of dynamic cells probed by a query. Computation is free in the locality model, cell contents may be arbitrary nonlinear functions of the source state, and query probes may be fully adaptive.

The regime of interest is

$$M = \Theta(n), \qquad w = \Theta(\log n).$$

Here one word has enough information capacity to hold an entire answer, so a large query lower bound cannot be attributed merely to output size. The question is instead whether a low-write representation can arrange information so that every prescribed w-bit answer is concentrated in a few words.

A tempting conjecture is that constant write locality should force $\Theta(w)$ recovery locality. At write locality one, variants of this intuition can often be made rigorous. At write locality two, however, the universal statement is false. A seven-coordinate Fano-plane gadget already admits a linear representation with two fixed write locations per source bit and two word probes per requested triple. We show that this is not only a finite curiosity: the gadget can be packed into an explicit infinite family with

$$M = n, \qquad w = \Theta(\log n), \qquad u = 2, \qquad q = 2.$$

Thus the correct question is structural rather than purely numerical.

> **Which fixed batch families admit constant recovery locality under two fixed writes, and which families force recovery locality linear in the batch size?**

Our main theorem answers this for random batch families under a canonical fixed-support representation. Suppose the $M = \Theta(n)$ batches are sampled independently and uniformly from the w-subsets of $[n]$, where $w = \Theta(\log n)$. Then with high probability every exact canonical encoding into w-bit cells, in which each source coordinate can affect at most two fixed physical cells, satisfies

$$\boxed{q = \Omega(w)}.$$

The theorem allows arbitrary nonlinear cell functions and fully adaptive exact decoding.

The proof exposes a geometric dichotomy in the fixed update-support system. Think of every source coordinate as a set of one or two memory cells. If the support system is **diffuse**, then a random requested batch contains many pairwise disjoint supports. An adaptive query must hit every one of them, immediately forcing $\Omega(w)$ probes. If the support system is **hub-concentrated**, a small set H of memory cells touches a constant fraction of the source coordinates. The hw bits stored in H can be given to the decoder for free as a sketch. Canonicality then limits how many source coordinates can be hidden entirely inside the hub and bounds the size of every residual nonhub source group. Random batches spread across those residual groups. After selecting several group-disjoint batches, exact recovery becomes a short-sketch problem for many independent target bits in distinct groups. A direct adaptive decision-tree counting argument then forces $\Omega(w)$ residual probes per original query.

The fixed-support hypothesis is essential to this geometry. If the locations modified by coordinate j may depend on the current state or update history, the static support graph disappears. To separate what is genuinely known from what remains open, we also prove a **state-local congestion inequality** that does not require fixed supports or a canonical global representation. At any reachable state, if a query uses q probes and each update changes at most u cells, then low q forces some current memory cell to be influenced by many source coordinates. For random batch families this implies that any hypothetical $q = o(w)$ moving-support scheme must maintain high-influence write hubs at every reachable state. The unrestricted $u = 2$ case is therefore reduced to the question of whether such hubs can move coherently enough to beat the fixed-support barrier.

1.1 Results

The paper establishes three principal results.

Explicit structured upper bound. We give an explicit Fano-based family with $M = n$, $w = \Theta(\log n)$, all source coordinates active, fixed write locality $u = 2$, and query locality $q = 2$. This proves that two-write constant recovery is asymptotically possible for highly structured incidence families.

State-local congestion barrier. For an arbitrary exact dynamic representation at a reachable state σ, let $f(\sigma)$ be the maximum number of source coordinates whose current one-bit update can modify the same cell. If $\rho(\mathcal{B})$ denotes the repeated-pair excess of the batch family, then

$$M\left(\frac{w^2}{q} - w\right) \le un(f(\sigma) - 1) + 2\rho(\mathcal{B}).$$

For a random w-uniform family with $M = \Theta(n)$ and $w = \Theta(\log n)$, $\rho(\mathcal{B}) = \mathrm{polylog}(n)$ with high probability. Hence bounded state-local congestion $f(\sigma) = O(w)$ already forces $q = \Omega(w)$, even without fixed support. Conversely, $q = o(w)$ requires

$$f(\sigma) = \Omega\left(\frac{w^2}{uq}\right)$$

at every reachable state.

Random-family fixed-support lower bound. For random $M = \Theta(n)$, $w = \Theta(\log n)$ batch families, every exact canonical nonlinear encoding with at most two fixed update locations per source coordinate has $q = \Omega(w)$ with high probability, even for fully adaptive queries.

These results yield a three-way frontier:

- **structured fixed-support** $u = 2$**:** $q = 2$;
- **random fixed-support** $u = 2$**:** $q = \Omega(w)$;
- **moving/history-dependent** $u = 2$**:** a high-congestion escape is required, while the full $\Omega(w)$ bound remains open.

1.2 Contribution boundary

The surrounding literature contains several closely related ingredients. Batch codes and combinatorial batch codes study retrieval from distributed storage (Ishai et al. 2004; Paterson, Stinson, and Wei 2009; Rawat et al. 2016). Locally updatable and locally decodable codes and compression schemes explicitly study simultaneous read and write locality (Chandran, Kanukurthi, and Ostrovsky 2014; Vatedka and Tchamkerten 2020; Vatedka, Chandar, and Tchamkerten 2020). Arbitrary-access-set storage codes tailor coding to prescribed access families and optimize update cost (Maturana and Rashmi 2021). Functional batch codes study recovery of prescribed linear combinations and, recently, bounded-size recovery sets (Y. Zhang, Etzion, and Yaakobi 2020; Oksner et al. 2026; Boruchovsky et al. 2026). Query-with-Sketch gives a general modern framework for adaptive probing with a short global sketch (Garg et al. 2026). Dynamic cell-probe and Multiphase lower bounds provide an important neighboring lower-bound tradition (Pătraşcu 2010; Larsen, Weinstein, and Yu 2020; Larsen and Yu 2026; Ko 2025, 2026a, 2026b).

Accordingly, we do **not** claim that local update plus local decode, two-replication support graphs, finite-geometry coding, sketch-aided adaptive probing, or the general dynamic cell-probe/Multiphase framework are new ideas. The bounded claim is narrower: within the literature searched through October 2026, we did not identify a materially equivalent theorem showing that a random fixed family of $\Theta(\log n)$-coordinate exact retrieval requests forces $\Omega(\log n)$ adaptive word probes under two **fixed** update locations per source coordinate for arbitrary nonlinear canonical encodings. We therefore describe the theorem as **likely distinct within the searched scope**, not as an absolute priority claim.

2. Model and notation

2.1 Fixed-Batch Dynamic Gather

An FBDG instance consists of a universe $[n] = \{1, \dots, n\}$ and a fixed batch family

$$\mathcal{B} = (B_1, \dots, B_M), \qquad B_i \in \binom{[n]}{w}.$$

The maintained logical state is $x \in \{0,1\}^n$. An update flips one coordinate x_j. Query i must return the exact ordered restriction $x|_{B_i}$.

Dynamic memory consists of cells that each store exactly w bits. The description of $\mathcal{B}$, the encoding/decoding algorithms, and state-independent constants are free. We count dynamic cell probes. A query may adapt its next address to values observed in earlier probes. Unless stated otherwise, queries are deterministic and exact; the lower bounds extend immediately to zero-error randomized queries with a worst-case probe bound by fixing the random tape.

The locality parameters are:

- u: maximum number of dynamic cells whose contents may change after one source-coordinate update;
- q: maximum number of dynamic cells probed by one query.

The paper studies locality rather than word-operation time. In particular, the positive construction proves $q = 2$ in the cell-probe/locality sense; it does not claim that every arbitrary packed output permutation can be assembled in $O(1)$ instructions on every standard word-RAM instruction set.

Cell-width normalization and robustness. We use the same symbol w for batch size and cell width deliberately. A query returns exactly w bits, so choosing w-bit cells removes a trivial output-capacity explanation for a large probe lower bound: information-theoretically, one cell is large enough to hold the entire answer. The lower-bound proof is robust to replacing the cell width by any $b = \Theta(w)$. The diffuse branch is independent of cell capacity. In the hub branch, the capacity bounds hw and $(h+1)w$ become hb and $(h+1)b$, the residual groups remain of size $O(w^2)$ because $h = O(w)$ and $b = O(w)$, and the free-sketch length becomes hb; choosing the constant in $L = \Theta(h)$ as a function of the upper constant in b/w preserves the final $q = \Omega(w)$ conclusion. We keep $b = w$ in the formal statement and proof to avoid a second size parameter. No claim is made here for cell widths growing asymptotically faster than w.

2.2 Canonical encodings and essential fixed supports

A **canonical encoding** is a map

$$E : \{0,1\}^n \longrightarrow (\{0,1\}^w)^{\mathcal{C}},$$

where $\mathcal{C}$ is an arbitrary set of dynamic cells and $E_c(x)$ denotes the content of cell c. Canonicality means that memory is a function of the current logical state alone, not of the update path used to reach that state.

For a logical coordinate j, define its **essential support**

$$S_j := \{c \in \mathcal{C} : \exists x \text{ such that } E_c(x) \neq E_c(x \oplus e_j)\}.$$

The canonical fixed-support condition with write locality two is

$$|S_j| \leq 2 \qquad \forall j \in [n].$$

Using essential supports removes dummy write locations: $c \in S_j$ exactly when coordinate j can actually influence cell c in some state.

Define the active source set

$$A := \bigcup_{i=1}^{M} B_i.$$

Only coordinates in A are semantically required to be recoverable.

2.3 State-local supports for unrestricted representations

For the congestion theorem we allow the representation to be noncanonical and the write locations to move. Fix any reachable memory state σ representing logical vector x. We assume the update transition from a given memory state under a specified source-bit flip is deterministic; equivalently, any update randomness is

regarded as fixed before defining the state-local influence sets. Let $U_j(\sigma)$ be the set of cells whose contents change when the next update flips coordinate j. We assume

$$|U_j(\sigma)| \le u.$$

Let $P_B(\sigma)$ denote the set of cells probed along the actual adaptive query path for batch B at state σ. Define

$$d_c(\sigma) := |\{j : c \in U_j(\sigma)\}|, \qquad f(\sigma) := \max_c d_c(\sigma).$$

Thus $f(\sigma)$ measures instantaneous write congestion or influence concentration.

For an unordered source pair $\{a, b\}$ define its batch codegree

$$\deg_{\mathcal{B}}(a, b) := |\{i : \{a, b\} \subseteq B_i\}|,$$

and define the **pair-repetition excess**

$$\rho(\mathcal{B}) := \sum_{\{a,b\}\subseteq[n]} (\deg_{\mathcal{B}}(a, b) - 1)_+.$$

3. Structured two-write families

We first show that $u = 2$ does not universally imply $q = \Omega(w)$, even asymptotically when $w = \Theta(\log n)$ and $M = n$.

3.1 A seven-coordinate Fano gadget

Consider source bits

$$U_0, U_1, U_2, L_1, L_2, R_1, R_2 \in \{0, 1\}.$$

Store three 3-bit cells

$$\begin{aligned} C_0 &= (U_0, L_1, L_2), \\ C_1 &= (U_1, R_1, R_2), \\ C_2 &= (U_2, L_1 \oplus R_1, L_2 \oplus R_2). \end{aligned}$$

Every U_i affects one cell. Each L_i affects C_0 and C_2, and each R_i affects C_1 and C_2. Therefore the fixed write locality is at most two.

Label the seven source coordinates by

$$1 = U_0, \quad 2 = U_1, \quad 3 = L_1, \quad 4 = U_2, \quad 5 = L_2, \quad 6 = R_1, \quad 7 = R_2.$$

Use the seven Fano lines

$$123,\ 145,\ 167,\ 246,\ 257,\ 347,\ 356.$$

Each line can be recovered from two cells. One fixed decoder assignment is shown below.

Requested line	Source bits	Two cells
123	U_0, U_1, L_1	C_0, C_1
145	U_0, U_2, L_2	C_0, C_2
167	U_0, R_1, R_2	C_0, C_1
246	U_1, U_2, R_1	C_1, C_2
257	U_1, L_2, R_2	C_0, C_1
347	L_1, U_2, R_2	C_0, C_2
356	L_1, L_2, R_1	C_0, C_1

For line 347, for example, C_0 reveals L_1, L_2 and C_2 reveals U_2 and $L_2 \oplus R_2$, so R_2 is recovered by one xor. The other cases are immediate or symmetric.

Proposition 3.1. The seven-line Fano family admits an exact linear canonical encoding with fixed write locality $u = 2$ and query locality $q = 2$.

The proposition was exhaustively revalidated computationally over all 2^7 logical states and all seven requested lines, for 896 state/query cases. A stronger check over every valid decoder-pair/line combination covered 1536 cases; see Appendix C.

3.2 Packing t Fano copies into three words

Take t independent copies of the seven-bit gadget. For copy $r \in [t]$, let $C_0^{(r)}, C_1^{(r)}, C_2^{(r)}$ be its three 3-bit encoded cells. Pack corresponding cells across copies into three global words

$$\widehat{C_k} := C_k^{(1)} \| C_k^{(2)} \| \cdots \| C_k^{(t)}, \qquad k \in \{0, 1, 2\}.$$

Each global word has

$$W := 3t$$

bits. Each source bit still affects at most two global words.

A global batch chooses one Fano line in each copy and asks for the union of the selected triples, so every global batch contains exactly $3t = W$ source bits. If every chosen line can be decoded from the same pair of global words, the whole batch is recoverable with two probes.

Define

$$\mathcal{L}_{01} = \{123, 167, 257, 356\}.$$

Every line in $\mathcal{L}_{01}$ avoids U_2 and is decodable from $(\widehat{C_0}, \widehat{C_1})$. Hence every sequence in $\mathcal{L}_{01}^t$ defines a distinct W-bit batch answerable from those two words. There are 4^t such sequences.

To ensure that every source coordinate is active, include the three global line patterns

$$(145)^t, \qquad (246)^t, \qquad (347)^t.$$

Together they cover all seven source labels in every copy. Complete the family by choosing any additional $7t - 3$ distinct patterns from $\mathcal{L}_{01}^t$. Since

$$4^t \geq 7t - 3 \qquad \text{for all } t \geq 1,$$

this gives exactly $7t$ distinct global batches, every one of size $W = 3t$, and every source coordinate is active.

Theorem 3.2 (explicit structured family). For every $t \geq 1$ there is an FBDG gadget with

$$n_g = M_g = 7t, \qquad w = 3t,$$

all source coordinates active, and an exact linear canonical encoding into three w-bit cells with

$$u = 2, \qquad q = 2.$$

Taking $g = 2^t$ disjoint gadgets yields

$$n = M = 7t2^t, \qquad w = 3t = \Theta(\log n), \qquad u = 2, \qquad q = 2.$$

Thus constant recovery locality persists asymptotically for a highly structured fixed batch family.

The construction is deliberately simple. Its purpose is not to optimize storage or coding rate against the batch-code literature, but to show that the two-write locality barrier is genuinely incidence-dependent.

4. A state-local congestion theorem

The next theorem does **not** assume fixed supports or a canonical global encoding. It identifies what any low-query representation must look like at each reachable state.

4.1 Adaptive hitting at a reachable state

Lemma 4.1 (state-local adaptive hitting). Fix a reachable memory state σ, a batch B, and $j \in B$. Then

$$P_B(\sigma) \cap U_j(\sigma) \neq \emptyset.$$

Proof. Let σ' be the memory state reached by applying the update that flips logical coordinate j. Suppose the query path on σ avoids every cell changed by that update. The first probe therefore sees the same value in σ and σ'. Inductively, because the adaptive algorithm sees the same transcript, it chooses the same next address, which is again outside the changed set and therefore has the same value. The entire transcript and output are identical on σ and σ', contradicting exactness because requested bit j changed. □

4.2 Congestion inequality

Theorem 4.2 (state-local congestion). Consider any exact FBDG representation at a reachable state σ. Suppose every one-bit update changes at most u cells and every query probes at most q cells. Then

$$\boxed{M\left(\frac{w^2}{q} - w\right) \leq un(f(\sigma) - 1) + 2\rho(\mathcal{B}).}$$

Equivalently,

$$q \geq \frac{w^2}{w + u(f(\sigma) - 1) + 2\rho(\mathcal{B})/M}.$$

Proof. Fix batch B_i. By Lemma 4.1, for every $j \in B_i$ choose one cell

$$\phi_i(j) \in P_{B_i}(\sigma) \cap U_j(\sigma).$$

For each probed cell c, let

$$m_{i,c} := |\{j \in B_i : \phi_i(j) = c\}|.$$

The nonzero $m_{i,c}$ form a partition of the w source coordinates among at most q cells. Hence, by Cauchy-Schwarz,

$$\sum_c m_{i,c}^2 \geq \frac{w^2}{q}.$$

Therefore the number of unordered source pairs assigned to a common cell satisfies

$$\sum_c \binom{m_{i,c}}{2} = \frac{1}{2}\left(\sum_c m_{i,c}^2 - w\right) \geq \frac{1}{2}\left(\frac{w^2}{q} - w\right).$$

Summing over all M batches gives at least

$$\frac{M}{2}\left(\frac{w^2}{q} - w\right)$$

pair/batch incidences whose two coordinates share a current update cell.

Now upper-bound those incidences. Ignoring multiplicity across batches, the number of distinct source pairs sharing at least one current update cell is at most

$$\sum_c \binom{d_c(\sigma)}{2} \leq \frac{f(\sigma) - 1}{2} \sum_c d_c(\sigma).$$

Since each of the n source coordinates changes at most u cells,

$$\sum_c d_c(\sigma) = \sum_{j=1}^{n} |U_j(\sigma)| \leq un.$$

A source pair that occurs in several batches can contribute at most one incidence without repetition plus at most

$$\deg_{\mathcal{B}}(a,b)-1$$

additional incidences. Summing these extras gives $\rho(\mathcal{B})$. Hence

$$\frac{M}{2}\left(\frac{w^2}{q}-w\right)\le\frac{un(f(\sigma)-1)}{2}+\rho(\mathcal{B}),$$

which proves the theorem after multiplying by two. □

4.3 Random families have negligible pair repetition

Lemma 4.3. Let $M=\Theta(n)$ and let the batches be independent uniform w-subsets with $w=\Theta(\log n)$. Then with high probability

$$\rho(\mathcal{B})=O(w^5)=o(n).$$

Proof. For a fixed pair $\{a,b\}$, let $D_{ab}=\deg_{\mathcal{B}}(a,b)$. Its success probability in one batch is

$$p_2=\frac{w(w-1)}{n(n-1)}.$$

Since $(D-1)_+\le\binom{D}{2}$,

$$\mathbb{E}\rho\le\binom{n}{2}\binom{M}{2}p_2^2=O(w^4).$$

Markov's inequality gives

$$\Pr[\rho>w^5]=O(1/w)=o(1).$$

□

Corollary 4.4 (bounded congestion is hard). For such a random family, if $u=O(1)$ and

$$f(\sigma)=O(w)$$

at a reachable state, then

$$q=\Omega(w).$$

Corollary 4.5 (moving-hub requirement). For such a random family, any exact scheme with $u=O(1)$ and $q=o(w)$ must satisfy, at every reachable state,

$$f(\sigma)\ge 1+\frac{w^2/q-w-o(1)}{u}=\Omega\left(\frac{w^2}{uq}\right).$$

In particular, at $q=\Theta(w/\log w)$ and constant u, some current cell must be influenced by $\Omega(w\log w)$ source coordinates.

The corollary is the structural reason the moving-support case is difficult: any escape from the fixed-support lower bound must continually create and move highly congested write hubs.

5. Main lower bound for random fixed-support families

We now prove the paper's main theorem.

Theorem 5.1 (random fixed-support nonlinear two-write barrier). Fix constants

$$0<c_-\le c_+<\infty,\qquad 0<\mu_-\le\mu_+<\infty.$$

Suppose

$$c_- \log n \le w \le c_+ \log n, \qquad \mu_- n \le M \le \mu_+ n.$$

Choose $B_1, \dots, B_M$ independently and uniformly from $\binom{[n]}{w}$. Then there exists a constant

$$c_* = c_*(c_-, c_+, \mu_-, \mu_+) > 0$$

such that, with probability $1 - o(1)$ over $\mathcal{B}$, every exact canonical FBDG encoding into w-bit cells satisfying

$$|S_j| \le 2 \qquad \forall j$$

has worst-case adaptive query locality

$$q \ge c_* w.$$

Thus

$$\boxed{q = \Omega(w)}.$$

No linearity assumption is made on the cell functions.

The proof is uniform over **all** canonical fixed-support encodings: the random family is sampled first, and with high probability it is hard simultaneously for every possible support system and every nonlinear choice of cell contents.

5.1 Parameters and proof roadmap

Set

$$a := \frac{1}{8}.$$

Choose a sufficiently small constant $\varepsilon \in (0, 1/16)$ such that

$$\Gamma := (1 - a) \log \frac{1}{2\varepsilon} - \log 2$$

satisfies

$$\Gamma \mu_- c_- > 3.$$

Define

$$\delta := \frac{\varepsilon}{2}, \qquad \kappa := \frac{\delta}{4} = \frac{\varepsilon}{8}, \qquad r := \lfloor aw \rfloor.$$

All logarithms in probability estimates are natural unless a base is explicitly written.

The proof first establishes five random-family events uniformly: almost all source coordinates are active; every source coordinate has batch degree $O(w)$; every linear-size source set appears substantially in many batches; every polylogarithmic source set has bounded repeated clustering; and every diffuse fixed support system has some batch containing a linear-size disjoint support matching. We then fix an arbitrary support system. Either it is diffuse and the matching directly forces $\Omega(w)$ probes, or it has a small hub. The hub case is reduced to a short-sketch exact-recovery problem on small residual groups.

6. Deterministic consequences of canonical fixed support

6.1 Cell-local dependence

For a cell c define

$$N(c) := \{j : c \in S_j\}.$$

Lemma 6.1 (cell-local dependence). If $x, y \in \{0,1\}^n$ agree on $N(c)$, then

$$E_c(x) = E_c(y).$$

Hence cell c is a function only of $x|_{N(c)}$.

Proof. Transform x into y by flipping only coordinates outside $N(c)$. By definition of essential support, none of those flips can change cell c. Canonicality identifies every intermediate memory state with the encoding of the intermediate logical vector, so the content of c remains unchanged throughout the path. □

This is the first point at which fixed support and canonicality become essential. With history-dependent memory states or state-dependent support locations, the same static dependency set $N(c)$ need not exist.

6.2 Active-coordinate injectivity

Lemma 6.2. If two source vectors x, y differ on some active coordinate $j \in A$, then

$$E(x) \neq E(y).$$

Proof. Some requested batch contains j. If $E(x) = E(y)$, the exact query algorithm would receive identical dynamic memory for both states and therefore return the same answer, although the requested vectors differ in coordinate j. □

6.3 Adaptive hitting

Lemma 6.3 (fixed-support adaptive hitting). Fix x, a requested batch B, and let $P_B(x)$ be the set of cells probed on the actual adaptive execution path. Then for every $j \in B$,

$$P_B(x) \cap S_j \neq \emptyset.$$

Proof. Suppose the path avoids S_j. The encodings $E(x)$ and $E(x \oplus e_j)$ differ only inside S_j. The first cell read on the run from x therefore has the same value in both states. Inductively, after identical transcripts the adaptive decoder chooses the same next address; that address lies on the original path, hence outside S_j, and again returns the same value. The complete transcript and output coincide, contradicting exactness because requested bit j changed. □

Corollary 6.4. Every actual probe set is a hitting set for $\{S_j : j \in B\}$. If a batch contains r coordinates whose supports are pairwise disjoint, then $q \geq r$.

7. Random-family events

All events in this section hold simultaneously with probability $1 - o(1)$.

7.1 Almost all coordinates are active

Lemma 7.1. With high probability,

$$|[n] \setminus A| \leq \frac{\varepsilon n}{4}.$$

Proof. For fixed j,

$$\Pr[j \notin A] = \left(1 - \frac{w}{n}\right)^M \leq \exp(-Mw/n) \leq \exp(-\mu_- w) \leq n^{-\mu_- c_-}.$$

The expected number of inactive coordinates is therefore $o(n)$. Markov's inequality gives the stated linear bound. □

7.2 Uniform source-degree bound

Let

$$d_j := |\{i : j \in B_i\}|.$$

Lemma 7.2. There exists a constant $\Delta = \Delta(c_-, \mu_+)$ such that with high probability

$$d_j \le \Delta w \qquad \forall j \in [n].$$

Proof. For fixed j, d_j is binomial with mean at most $\mu_+ w$. A Chernoff bound gives

$$\Pr[d_j \ge \Delta w] \le \Big(\frac{e\mu_+}{\Delta}\Big)^{\Delta w}.$$

Choose Δ large enough that the right side is at most n^{-3} for every $w \ge c_- \log n$, then union bound over the n coordinates. □

7.3 Uniform richness of linear-size sets

Lemma 7.3. With high probability, every $C \subseteq [n]$ with $|C| \ge \delta n$ has at least $M/2$ batches satisfying

$$|B_i \cap C| \ge \frac{\delta w}{2}.$$

Proof. Fix C. For one batch, $X = |B_i \cap C|$ is hypergeometric with mean at least δw. The usual hypergeometric lower-tail Chernoff inequality gives

$$\Pr[X < \delta w/2] \le e^{-\delta w/8} =: p.$$

The M batches are independent. The probability that at least $M/2$ batches are bad is at most

$$2^M p^{M/2} \le \exp\left(M \log 2 - \frac{\delta M w}{16}\right) = \exp(-\Omega(nw)).$$

There are at most 2^n choices of C, and $w = \Theta(\log n)$, so a union bound remains $o(1)$. □

7.4 Uniform spreading on polylogarithmic sets

Define

$$D_0 := (2aw + 1)w = O(w^2).$$

For $S \subseteq [n]$, let

$$Z(S) := \sum_{i=1}^{M} (|B_i \cap S| - 1)_+.$$

This quantity counts repeat incidences beyond the first hit of S in each batch.

Lemma 7.4 (uniform spreading). There is a constant $K \ge 3$ such that with high probability, simultaneously for every S with $1 \le |S| \le D_0$,

$$\boxed{Z(S) \le K|S|.}$$

Proof. Fix S of size $s \le D_0$ and put $t = \lceil Ks \rceil$. If $Z(S) \ge t$, then for some $1 \le r' \le t$ there are distinct batches $i_1, \dots, i_{r'}$ and positive integers $t_1, \dots, t_{r'}$ summing to t such that

$$|B_{i_\ell} \cap S| \ge t_\ell + 1 \qquad (\ell = 1, \dots, r').$$

For any specified k-subset $T \subseteq S$,

$$\Pr[T \subseteq B_i] = \frac{(w)_k}{(n)_k} \le \Big(\frac{w}{n}\Big)^k.$$

For fixed r' and composition $(t_1, \dots, t_{r'})$, a union bound over selected batches and selected source subsets therefore gives at most

$$\binom{M}{r}' \prod_{\ell=1}^{r'} \binom{s}{t_\ell + 1} \left(\frac{w}{n}\right)^{t_\ell+1}.$$

Using $\binom{s}{k} \le s^k$ and summing over the at most 2^{t-1} compositions of t,

$$\Pr[Z(S) \ge t] \le 2^t \sum_{r'=1}^{t} M^{r'} \left(\frac{sw}{n}\right)^{t+r'}.$$

Since $M \le \mu_+ n$ and $r' \le t$,

$$M^{r'} \left(\frac{sw}{n}\right)^{t+r'} = \left(\frac{sw}{n}\right)^{t} (\mu_+ sw)^{r'} \le \left(\frac{Cs^2w^2}{n}\right)^{t}$$

for a constant C after absorbing the sum over r'. Thus

$$\Pr[Z(S) \ge Ks] \le \left(\frac{C's^2w^2}{n}\right)^{Ks}.$$

Because $s \le D_0 = O(w^2)$, the base is at most $C''w^6/n$. Union bounding over the at most $(en/s)^s$ sets of size s gives

$$\Pr[\exists S, |S| = s : Z(S) > Ks] \le \left[\frac{en}{s}\left(\frac{C''w^6}{n}\right)^{K}\right]^{s}.$$

For any fixed $K \ge 3$, the bracket is $n^{1-K}\,\mathrm{polylog}(n) = o(1)$. Summing over $1 \le s \le D_0 = O(\log^2 n)$ proves the lemma. □

7.5 A uniform matching event for every diffuse support system

Call a support system **diffuse** if every cell set T with

$$|T| \le 2r$$

is incident to at most εn source coordinates:

$$|\{j : S_j \cap T \ne \emptyset\}| \le \varepsilon n.$$

Lemma 7.5 (uniform diffuse matching). With high probability over $\mathcal{B}$, every diffuse fixed support system contains some requested batch with $r = \lfloor aw \rfloor$ pairwise disjoint supports.

Proof. First count relevant support systems. The union of all essential supports contains at most $2n$ cells. A cell belonging to no support is constant over the Boolean cube by Lemma 6.1 and can be compiled into the query algorithm. Relabel the at-most-$2n$ support cells by $[2n]$. Each source coordinate chooses a subset of size 0, 1, or 2, so the number of labeled support systems is at most

$$\left(1 + 2n + \binom{2n}{2}\right)^n \le (3n^2)^n = \exp(2n \log n + O(n)).$$

Fix one diffuse support system. Expose one random batch as an ordered sample of w distinct source coordinates. Greedily accept a coordinate if its support is disjoint from all supports previously accepted. Before r acceptances, the union T of accepted supports has size at most $2r$, so at most εn source coordinates conflict with T. Conditional on the past, the next sampled coordinate is therefore rejected with probability at most

$$\frac{\varepsilon n}{n - w} \le 2\varepsilon$$

for sufficiently large n.

If fewer than r coordinates are accepted, at least $w-r+1 \geq (1-a)w$ draws were rejected. Union bounding over rejection-position sets gives

$$\Pr[\text{matching size} < r] \leq 2^w (2\varepsilon)^{(1-a)w} = e^{-\Gamma w}.$$

All M batches fail with probability at most

$$e^{-\Gamma M w} \leq \exp(-\Gamma \mu_c_ n \log n).$$

Finally, union bound over every labeled support system. The total failure probability is at most

$$\exp\left((2 - \Gamma \mu_c_ + o(1)) n \log n\right) = o(1)$$

by the choice $\Gamma \mu_c_ > 3$. □

8. An adaptive sketch-probe lemma

The hub case can be closed with a short self-contained counting lemma. It is a specialized exact-product analogue of the broader Query-with-Sketch methodology of Garg et al. (Garg et al. 2026).

Lemma 8.1 (independent-target sketch-probe bound). Let

$$X = (X_1, \dots, X_N)$$

be independent group-valued random variables. Let

$$F(X) = (Y_1, \dots, Y_m) \in \{0,1\}^m,$$

where the Y_ℓ are independent unbiased bits and each target bit belongs to a distinct input group. Consider any deterministic algorithm that

1. receives an arbitrary s-bit sketch $S(X)$;
2. adaptively probes whole input groups X_i;
3. computes $F(X)$ exactly.

For every integer $c \geq 1$, if $m > s + c$, its average number D of group probes satisfies

$$\boxed{\mathbb{E}[D] \geq (1 - 2^{-c})(m - s - c).}$$

In particular,

$$\mathbb{E}[D] \geq \frac{3}{4}(m - s - 2)$$

for $c = 2$.

Proof. For each sketch value $z \in \{0,1\}^s$, fix the adaptive decision tree used when the sketch equals z. Put

$$t := m - s - c.$$

Consider a leaf reached after at most t group probes, with probe transcript σ and output $y \in \{0,1\}^m$. At most t target groups have been revealed, so conditioned only on σ, at least

$$m - t = s + c$$

target bits remain independent and unbiased. Therefore

$$\Pr[F(X) = y \mid \sigma] \leq 2^{-(s+c)}.$$

For a fixed sketch tree, the cylinder events σ associated with distinct leaves are disjoint. Hence the total probability mass of inputs that both reach that tree, halt within t probes, and are correct is at most $2^{-(s+c)}$. Summing over the at most 2^s sketch values gives

$$\Pr[\text{correct and } D \le t] \le 2^{-c}.$$

The algorithm is exact, so

$$\Pr[D > t] \ge 1 - 2^{-c}.$$

Consequently

$$\mathbb{E}[D] \ge t \Pr[D > t] \ge (1 - 2^{-c})(m - s - c).$$

□

Two details are worth emphasizing. First, adaptivity is fully allowed: each sketch value defines an arbitrary decision tree. Second, the proof never conditions on the sketch value when claiming residual entropy; that would be invalid for a maliciously informative sketch. Instead, the 2^s possible sketch values are paid for by the final union bound.

9. Proof of the main theorem

Assume the random family simultaneously satisfies Lemmas 7.1–7.5. Fix an arbitrary canonical fixed-support encoding with $|S_j| \le 2$. We split into two cases.

9.1 Diffuse case

Suppose the support system is diffuse. By Lemma 7.5, some requested batch contains

$$r = \lfloor aw \rfloor$$

pairwise disjoint supports. By Lemma 6.3, the actual adaptive query path must hit every one of those supports. A single probed cell can hit at most one support in a pairwise disjoint family. Therefore

$$q \ge r \ge \frac{a}{2} w$$

for all sufficiently large n.

9.2 Hub-concentrated case

Otherwise there is a cell set

$$H \subseteq \mathcal{C}, \qquad h := |H| \le 2r \le 2aw,$$

incident to more than εn source coordinates.

Coordinates internal to the hub

Let

$$I_H := \{j \in A : S_j \subseteq H\}.$$

Lemma 9.1.

$$|I_H| \le hw.$$

Proof. Fix all logical coordinates outside I_H and vary the bits in I_H. By Lemma 6.1, only cells in H can change. By active-coordinate injectivity, all $2^{|I_H|}$ assignments to I_H must yield distinct hub contents. The hub contains only hw bits, so

$$2^{|I_H|} \le 2^{hw}.$$

□

Since $h = O(w)$ and $w = \Theta(\log n)$,

$$hw = O(w^2) = o(n).$$

Lemma 7.1 leaves at most $\varepsilon n/4$ inactive coordinates. Therefore, for large n, at least

$$\varepsilon n - \frac{\varepsilon n}{4} - hw \geq \frac{\varepsilon n}{2} = \delta n$$

active coordinates incident to H have an essential support with exactly one endpoint in H and one outside. Let C denote this crossing set.

Residual source groups

For every nonhub cell $v \notin H$, define

$$G_v := \{j \in C : v \in S_j\}.$$

The nonempty groups G_v partition C.

Lemma 9.2. For every $v \notin H$,

$$|G_v| \leq (h+1)w \leq D_0.$$

Proof. Fix all source coordinates outside G_v. By fixed-support locality, varying G_v can affect only the hub cells and cell v. Active-coordinate injectivity therefore embeds $2^{|G_v|}$ distinct assignments into at most $(h+1)w$ bits:

$$2^{|G_v|} \leq 2^{(h+1)w}.$$

The second inequality follows from $h \leq 2aw$. □

Many batches touch many residual groups

For a batch B_i, let

$$r_i := |B_i \cap C|$$

and define its residual collision excess

$$e_i := \sum_v (|B_i \cap G_v| - 1)_+.$$

The number g_i of distinct residual groups touched by B_i satisfies

$$g_i = r_i - e_i.$$

By Lemma 7.3, at least $M/2$ batches have

$$r_i \geq \frac{\delta w}{2}.$$

By Lemma 7.4 applied to every G_v and the fact that the groups partition C,

$$\sum_i e_i = \sum_v Z(G_v) \leq K \sum_v |G_v| = K|C| \leq Kn.$$

Thus only

$$\frac{4Kn}{\delta w} = o(n)$$

batches can have $e_i > \delta w/4$. Since $M = \Theta(n)$, a constant fraction of all batches satisfy both inequalities, and every such batch touches at least

$$\frac{\delta w}{4} = \kappa w$$

distinct residual groups.

Selecting group-disjoint batches

By Lemma 7.2 and Lemma 9.2, any fixed group G_v is touched by at most

$$\sum_{j \in G_v} d_j \le \Delta w |G_v| \le \Delta(h+1)w^2$$

batches. A candidate batch touches at most w residual groups, so selecting one candidate can conflict with at most

$$\Delta(h+1)w^3$$

other candidates.

Set

$$L := \left\lceil \frac{4h}{\kappa} \right\rceil.$$

Since $h = O(w)$,

$$L\,\Delta(h+1)w^3 = O(w^5) = o(n).$$

The candidate pool has $\Theta(n)$ batches. A greedy procedure therefore selects L candidate batches whose touched residual-group sets are pairwise disjoint.

Each selected batch touches at least κw groups. Choose one requested source bit from each touched group. Across all selected batches this gives

$$m \ge \kappa L w$$

target bits, each in a different residual group.

Reducing the hub case to sketch-probe recovery

Consider the product distribution in which all coordinates outside C are fixed to zero and all bits in C are independent unbiased bits. For every residual group define the raw group variable

$$X_v := x|_{G_v}.$$

The X_v are independent.

Give the decoder the complete contents of all h hub cells for free. This is an arbitrary sketch of length

$$s := hw.$$

Whenever an original FBDG query probes a nonhub cell v with $G_v \neq \emptyset$, reveal the complete raw group X_v. By Lemma 6.1 and because every source bit outside C is fixed, the encoded cell $E_v(x)$ is a deterministic function of X_v and fixed constants, so one raw-group probe can simulate the original cell probe. If $G_v = \emptyset$, the cell is constant under the hard distribution and may be answered for free. Hub probes are answered from the sketch.

Run the L selected FBDG queries, caching any raw group once revealed. The resulting exact sketch-probe algorithm recovers the m selected target bits using at most

$$Lq$$

raw-group probes. The targets are independent unbiased bits and occupy distinct groups, so Lemma 8.1 with $c = 2$ gives

$$Lq \ge \frac{3}{4}(m - s - 2).$$

Now

$$m \ge \kappa L w$$

and, by the choice of L,

$$\frac{s}{L} = \frac{hw}{L} \le \frac{\kappa w}{4}.$$

Therefore

$$q \geq \frac{3}{4}\left(\kappa w - \frac{hw}{L} - \frac{2}{L}\right) \geq \frac{9\kappa}{16}w - o(1).$$

For sufficiently large n this is at least $\kappa w/2$. Combining the diffuse and hub cases proves Theorem 5.1 with, for example,

$$c_* := \min\left\{\frac{a}{2}, \frac{\kappa}{2}\right\} > 0.$$

□

10. Why fixed support is the exact proof boundary

The main theorem is sometimes easy to overread as a general dynamic lower bound. It is not. Fixed support is used in four logically distinct places.

First, Lemma 6.1 requires a static set $N(c)$ of source coordinates that can ever affect cell c. Second, the diffuse branch requires one static support system over which a random-family matching event can be union-bounded. Third, the hub branch needs a permanent set H and a permanent partition into residual groups G_v. Fourth, simulation of a nonhub encoded cell by one raw residual group relies on the statement that no source coordinate outside G_v can alter that cell under the hard distribution.

All four can fail when update locations move. Even a tiny canonical example can exhibit support motion. Suppose two logical bits x_1, x_2 are stored in two 2-bit cells A, B so that

$$\begin{aligned} x_2 = 0: &\quad A = (x_1, 0), \quad B = (0, 0), \\ x_2 = 1: &\quad A = (0, 0), \quad B = (x_1, 1). \end{aligned}$$

Flipping x_1 changes A when $x_2 = 0$ and B when $x_2 = 1$; flipping x_2 changes at most the two cells. Thus $u \leq 2$, yet x_1 has no single fixed physical support across all states.

This example does not beat any lower bound; it only demonstrates that static residual-group geometry cannot be assumed in the moving-support model. The congestion theorem of Section 4 is therefore important: it remains valid without fixed support and proves that any $q = o(w)$ moving-support construction for a random family must replace static geometry by continually moving high-influence hubs.

11. Related work

11.1 Batch and combinatorial batch codes

Batch codes were introduced by Ishai, Kushilevitz, Ostrovsky, and Sahai to support simultaneous retrieval with bounded access to each bucket (Ishai et al. 2004). Combinatorial batch codes restrict storage to replication and lead naturally to incidence and Hall-type formulations (Paterson, Stinson, and Wei 2009). Graph-based constructions, including dense high-girth graphs, provide an extensive structural language for low-replication batch retrieval (Rawat et al. 2016). Restricted reconstruction/query size has also been studied directly in batch-code models (H. Zhang and Skachek 2015; Thomas and Skachek 2017), while later work strengthened redundancy lower bounds for linear batch codes (Li and Wootters 2021). In a two-replication combinatorial batch code, an item is naturally represented by an edge joining its two storage locations, which is closely related to our support-system picture.

The objective is different here. FBDG fixes a particular query family in advance, allows arbitrary nonlinear encoded cell contents, and couples retrieval to dynamic point updates. The lower bound is not a storage-rate bound and does not require disjoint recovery sets for simultaneous requests. We therefore treat the support graph as established surrounding machinery rather than a novelty claim.

11.2 Functional batch codes and small recovery sets

Functional PIR and batch codes allow queries for linear combinations of information symbols and study the number of servers needed for multiple disjoint recovery sets (Y. Zhang, Etzion, and Yaakobi 2020). Recent work explicitly studies functional batch codes in which each query must be answered from a small number of coded symbols (Oksner et al. 2026), and all-symbol PIR/batch variants continue to refine recovery structure (Boruchovsky et al. 2026). These works are especially relevant to the positive side of our paper because they show that algebraic structure can make many prescribed recoveries local.

The main lower bound here differs in three dimensions: the requested objects are exact source-coordinate batches from one fixed random family; the representation evolves under point updates with a hard cap on write locations; and the encoding may be nonlinear. We did not find a theorem in this literature with the same random-family, two-fixed-write, nonlinear adaptive-recovery statement.

11.3 Local update and local decode

Update efficiency in coding theory predates the specific locally-updatable/locally-decodable formulations: Mazumdar, Chandar, and Wornell study how many encoded symbols must change after a single information-symbol update in capacity-approaching codes (Mazumdar, Chandar, and Wornell 2013). Chandran, Kanukurthi, and Ostrovsky introduced locally updatable and locally decodable codes, showing explicitly that low write and read locality can coexist under suitable coding metrics (Chandran, Kanukurthi, and Ostrovsky 2014). Vatedka and Tchamkerten studied compression with local decoding and local update, including constant average locality (Vatedka and Tchamkerten 2020). Vatedka, Chandar, and Tchamkerten obtained $O(\log \log n)$ worst-case local decoding and update efficiency together with converse results under stated assumptions (Vatedka, Chandar, and Tchamkerten 2020). These results make clear that simultaneous read/write locality is an established research direction.

FBDG removes noise tolerance and compression objectives and instead asks for exact recovery of one member of a fixed multi-coordinate query family. That specialization exposes an incidence parameter absent from single-coordinate local decoding: the same two-write representation may be constant-query for one family and linear-query for another.

11.4 Arbitrary access sets

Maturana and Rashmi study array codes tailored to prescribed storage-node access sets, with explicit attention to update cost and storage overhead (Maturana and Rashmi 2021). This is close in spirit to the thesis that workload structure matters. Their access sets specify subsets of storage nodes sufficient to reconstruct a message, whereas FBDG batches specify subsets of source coordinates that must be recovered from encoded dynamic memory. The orientations are different, but the comparison reinforces why the Fano/finite-geometry side of our paper should be viewed as part of an existing coding neighborhood rather than as a standalone coding novelty.

11.5 Query-with-Sketch

Garg, He, Li, Papakonstantinou, and Yang develop the Query-with-Sketch model for systematic data-structure lower bounds, allowing an arbitrary short sketch followed by adaptive probes and using conditional min-entropy to lift no-sketch lower bounds (Garg et al. 2026). The hub case in our proof has the same high-level architecture: hub cells become a sketch and residual encoded cells are strengthened to raw source-group probes. Because our hard distribution is a direct product and the target bits lie in distinct groups, the specialized Lemma 8.1 gives a shorter self-contained argument. We make no claim that sketch-aided decision-tree counting is new in itself.

11.6 Dynamic cell-probe and Multiphase lower bounds

Pătraşcu's Multiphase problem was introduced as a central vehicle for dynamic lower bounds (Pătraşcu 2010). Subsequent work pushed unconditional dynamic cell-probe lower bounds beyond the logarithmic barrier for

Boolean problems (Larsen, Weinstein, and Yu 2020) and for natural dynamic graph problems (Larsen and Yu 2026). More recent work develops a unified Multiphase/communication framework (Ko 2025), obtains an $\Omega((\log n/\log\log n)^2)$ lower bound for a Boolean Multiphase instantiation (Ko 2026a), and transfers the same scale to incremental graph problems including undirected shortest paths (Ko 2026b).

FBDG lies next to this lineage but asks a different question. Classical dynamic cell-probe results measure update and query **time** for a dynamic problem, usually with unrestricted locations for the cells modified by an update. FBDG instead fixes a family of vector-valued retrieval requests and asks what exact recovery locality is forced by an exceptionally strong write-location restriction: each source coordinate may affect at most two fixed essential cells. Conversely, Theorem 5.1 does not imply the stronger time lower bounds obtained in Multiphase-style work. The two lines are therefore complementary rather than competing: Multiphase results characterize general dynamic hardness, whereas the FBDG separation isolates how prescribed incidence structure interacts with fixed write locality.

11.7 Novelty boundary after the October 2026 search

The literature search for this manuscript included batch codes, graph batch codes, functional batch/PIR codes, arbitrary-access-set codes, locally updatable/locally decodable codes, local decode/update compression, adaptive sketch-probe lower bounds, write locality, the Multiphase framework, and modern dynamic cell-probe lower bounds. Recent 2026 work on functional batch codes with small recovery sets and on near-quadratic-logarithmic dynamic cell-probe lower bounds was included explicitly (Oksner et al. 2026; Ko 2026a, 2026b). No materially equivalent statement to Theorem 5.1 was found in that search.

This is evidence for bounded distinctness, not a proof of global priority. Accordingly, the manuscript avoids the words “first,” “optimal,” and “state of the art” as absolute claims.

12. Discussion

12.1 What the separation says

The positive and negative results share the same asymptotic batch size and the same write-locality budget. What changes is the incidence structure of the requested family. The structured construction deliberately aligns many requested triples with a small algebraic set of two-word decoders. Random families destroy this alignment in two complementary ways. If supports remain spread out, random batches expose many disjoint dependencies. If supports concentrate, random batches spread requested coordinates across many residual source groups after the concentrated cells are removed.

This provides a sharper explanation than the slogan “random queries are hard.” The actual obstruction is that a two-write canonical encoding has only two ways to make many coordinates jointly accessible: reuse the same cells heavily or keep their supports dispersed. The proof attacks those two possibilities separately.

12.2 Why nonlinearity does not rescue fixed supports

The lower bound never linearizes the cell contents. In the diffuse branch only the locations of essential dependencies matter. In the hub branch the capacity arguments use injectivity and bit counts, while the final simulation reveals complete raw residual groups, which is strictly more information than the original nonlinear encoded cell. Thus arbitrary nonlinear preprocessing inside each fixed support cell does not bypass the argument.

This is important because linear sparse-factorization lower bounds alone cannot establish the theorem: the structured Fano gadget itself is linear, but the hard-family theorem must exclude nonlinear alternatives as well.

12.3 Why adaptivity does not rescue fixed supports

Adaptivity is handled twice. The hitting lemma follows the actual decision path and shows that avoiding a changed support yields the same transcript after a source-bit flip. The sketch-probe lemma treats each sketch value as an arbitrary adaptive decision tree. No step replaces the adaptive decoder by a fixed nonadaptive probe set.

12.4 Toward deterministic hard families

Randomness is used only to guarantee a small list of combinatorial properties: almost-full activity, bounded coordinate degree, richness of every linear-size subset, spreading on polylogarithmic subsets, and a uniform diffuse matching event. This suggests a derandomization program. An explicit family satisfying deterministic analogues of these properties would turn Theorem 5.1 into an explicit hard-family result.

The most demanding requirement is the uniform diffuse matching property, because it must hold against every possible two-support system after the family is fixed. Expander- or extractor-like incidence constructions are natural candidates, but the required quantifier structure is stronger than ordinary pairwise expansion.

13. Limitations and open problems

The main theorem has deliberately narrow scope.

Moving support. The result does not cover state-dependent or history-dependent update locations. Section 4 shows that any sublinear-query escape must maintain high-congestion hubs, but it does not rule out moving hubs.

Canonicality. The fixed-support theorem assumes the memory representation is a function of the current source state. Canonicality is used in the cell-local dependence lemma. The state-local congestion theorem does not need this global assumption, but the $\Omega(w)$ fixed-support proof does.

Cell width. The formal model normalizes cell width to the batch size w. As noted in Section 2.1, the same asymptotic lower bound is robust to constant-factor cell widths $b = \Theta(w)$ after adjusting constants in the hub argument. Widths growing asymptotically faster than the answer size are outside the stated result.

Word-operation time. The paper is a locality/cell-probe result. The structured construction guarantees two cell reads, not a universal constant instruction count for assembling every packed output under every standard word-RAM instruction set.

Random rather than explicit hard family. The hard family is sampled randomly. Producing a deterministic explicit family with the same universal hardness property is open.

Constant optimization. The proof intentionally uses wasteful constants. The scientific statement is asymptotic $\Omega(w)$; improving c_* is not pursued.

These limitations lead to several concrete questions.

1. Can one prove $q = \Omega(w)$ for random w-uniform families with $u = 2$ when write locations may move with the state or history?
2. Does an analogous fixed-support theorem hold for every constant $u > 2$?
3. What deterministic incidence parameter exactly separates families admitting $q = O(1)$ from families forcing $q = \Omega(w)$?
4. Can the random-family events in Section 7 be realized by an explicit efficiently constructible family?
5. Can one characterize the additional power of noncanonical history-dependent representations separately from state-dependent physical support motion?

14. Conclusion

Fixed-batch dynamic retrieval exhibits a genuine structure-versus-randomness locality separation. Two fixed writes per source coordinate are sufficient for two-probe exact recovery on explicit highly structured families with logarithmic batch size. The same write budget is insufficient for random logarithmic-size batch families: with high probability every exact canonical representation, even with arbitrary nonlinear cells and fully adaptive queries, requires a linear number of word probes.

The proof identifies the mechanism behind the separation. A fixed support system is either diffuse, in which case a random request exposes many disjoint dependencies, or hub-concentrated, in which case the hub supplies only a short global sketch and the remaining information is dispersed across small residual groups. Random incidence defeats both geometries.

The remaining frontier is correspondingly precise. To beat the random-family barrier with two writes, an unrestricted representation must exploit support motion strongly enough to maintain high-influence hubs while changing which physical cells play that role across states. Whether such moving geometry can actually yield $q = o(w)$ is the central open problem.

Appendix A. Decoder classes for the Fano gadget

The three cell pairs can decode the following four Fano lines each:

$$\mathcal{L}_{01} = \{123, 167, 257, 356\},$$
$$\mathcal{L}_{02} = \{145, 167, 347, 356\},$$
$$\mathcal{L}_{12} = \{246, 257, 347, 356\}.$$

These sets follow from which unique U_i bit is unavailable from a given cell pair. For example, (C_0, C_1) directly reveals all source bits except U_2, so every Fano line avoiding point $4 = U_2$ is decodable from that pair.

The t-fold construction uses three fixed all-copy patterns to activate all source coordinates and fills the remaining query slots with distinct sequences from $\mathcal{L}_{01}^t$. Because $4^t \geq 7t - 3$, the construction has at least the required number of distinct patterns for every $t \geq 1$.

Appendix B. Theorem-dependency graph

```
Main random-family theorem
|
+-- canonical fixed-support model
|   +-- cell-local dependence
|   +-- active-coordinate injectivity
|   +-- adaptive hitting
|
+-- random-family events
|   +-- almost all coordinates active
|   +-- maximum source degree O(w)
|   +-- uniform richness of linear-size sets
|   +-- uniform spreading on O(w^2)-size sets
|   +-- uniform diffuse matching over every fixed support system
|
+-- diffuse/hub dichotomy
    |
    +-- diffuse
    |   +-- random batch contains Omega(w) disjoint supports
    |   +-- adaptive hitting
```

```
|   +-- q = Omega(w)
|
+-- hub
    +-- hub-internal active coordinates <= h w
    +-- Omega(n) active crossing coordinates
    +-- residual groups of size <= (h+1) w
    +-- many batches touch kappa w distinct groups
    +-- bounded source degree => L=Theta(h) group-disjoint batches
    +-- hub cells form an h w-bit sketch
    +-- nonhub cell probe <= one raw residual-group probe
    +-- independent-target sketch-probe lemma
    +-- L q >= Omega(L w)
    +-- q = Omega(w)
```

Appendix C. Reproducibility and validation notes

The general lower bounds are mathematical proofs, not empirical claims. The reproducibility artifact accompanying this manuscript therefore focuses on finite validation of the structured construction and on preserving the exact proof state.

The script `validation/fbdg_structured_validate.py` performs the following checks:

1. Exhaustive validation of the fixed decoder assignment for the seven Fano queries over all 2^7 states: 896/896 state/query cases pass.
2. Exhaustive validation of every valid cell-pair/line decoder relation over all 2^7 states: 1536/1536 cases pass.
3. Construction of $7t$ distinct all-active global batches for $t \in \{1, 2, 3, 4, 5, 8\}$.
4. Random differential checks of every constructed global query against raw source bits for each tested t.

Observed query-check counts in the retained run were:

t	batches	active source bits	randomized state/query checks
1	7	7	7,000
2	14	14	14,000
3	21	21	21,000
4	28	28	28,000
5	35	35	7,000
8	56	56	11,200

The exact retained stdout is stored in `validation/EXPECTED_OUTPUT.txt`. Its SHA-256 is `3f4cf123386c0239858e0ef597a91`
`validation/manifest.json` records the command, Python assumptions, expected counts, checksum, and validator Git-blob identity used when the reproducibility package was frozen on 2026-10-17. The repository CI reruns the validator and requires byte-for-byte agreement with the retained output.

These executions validate the finite implementation of the positive construction; they are not used as evidence for the asymptotic lower bound.

Appendix D. Proof-repair ledger

The final proof incorporates several defects found during adversarial auditing of earlier proof sketches.

1. **Spreading alone does not imply group-disjoint queries.** Two batches may touch the same residual group once each while contributing zero collision excess. The final proof adds the uniform source-degree bound and a greedy conflict argument.

2. **The number of aggregated batches needs an explicit constant.** A statement of the form $L = \Theta(h)$ is insufficient because the hub sketch can encode the entire target if the hidden constant is too small. The final proof sets $L = \lceil 4h/\kappa \rceil$.
3. **Inactive coordinates cannot be counted by injectivity.** A source bit never appearing in a query need not be represented. The proof first shows that only $o(n)$ random-family coordinates are inactive and applies information-capacity arguments only to active coordinates.
4. **Dummy support locations are excluded.** Essential supports are defined by actual state dependence.
5. **Adaptivity is preserved.** Both hitting and sketch-probe arguments work with actual adaptive decision trees.
6. **No fixed-to-moving support substitution occurs.** Every use of static support geometry is explicit, and Section 10 isolates the precise boundary.

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